

LogChirpy

MoCo SoSe 2025

Martin Lauterbach
Younna Samouneh
Luis Wehrberger



**Live Camera and
Callable for
Storage
ML Detection and
Classification**

**Bird Sighting
Storage**

**BirdyDex
Encyclopedia**

**30k+ Birds
with names in 6
languages, fully local
images and full
taxonomy facts**

Live Camera and
Callable for
Storage
ML Detection and
Classification

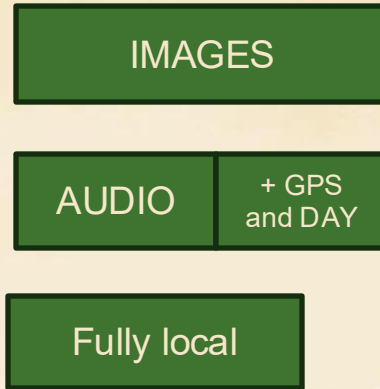
IMAGES

AUDIO

+ GPS
and DAY

Fully local

Live Camera and Callable for Storage ML Detection and Classification



Unified Pipeline in Live-Camera:

1. Take Photo
2. Object Detect
3. Crop per Obj
4. Classify per Obj
5. Save high-conf. to app storage
6. Display to User with CONF|LABEL
7. Record filtered audio for 3 seconds
8. decode and convert to right tensors
9. give to big audio model
 10. Take gps and day of year
 11. Give to small chance models
 12. Weigh output to big audio
10. model label results
13. Display to User
14. REPEAT



- Unified Pipeline Service (sequential processing)
- Audio classification with BirdNET models and fast-tensorflow package
- Object & Image detection with MLKit - Eliminates race conditions through sequential flow



Fully local ! Optional bird sighting firebase storage

Bird Sighting Storage

IMAGES

AUDIO

VIDEO

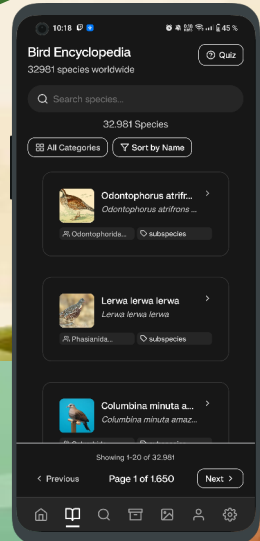
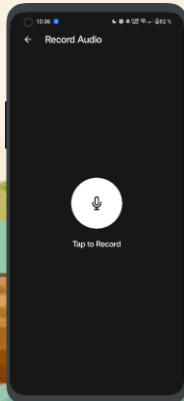
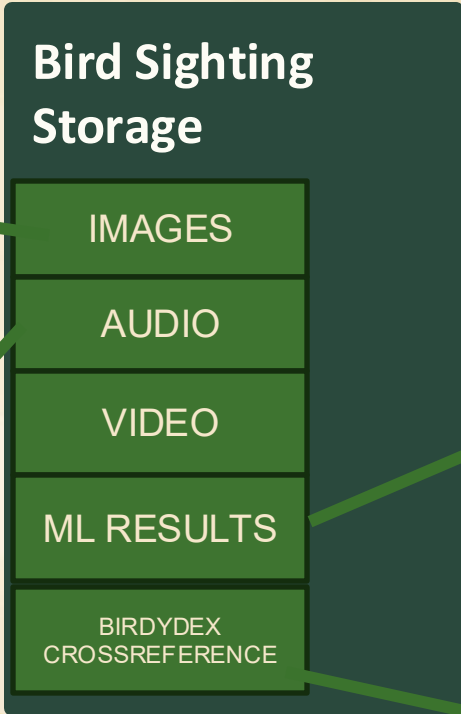
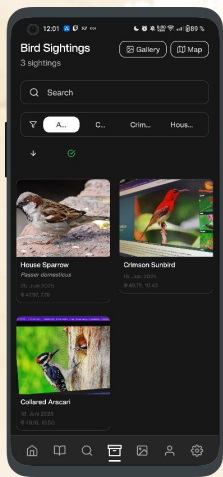
ML RESULTS

BIRDYDEX
CROSSREFERENCE

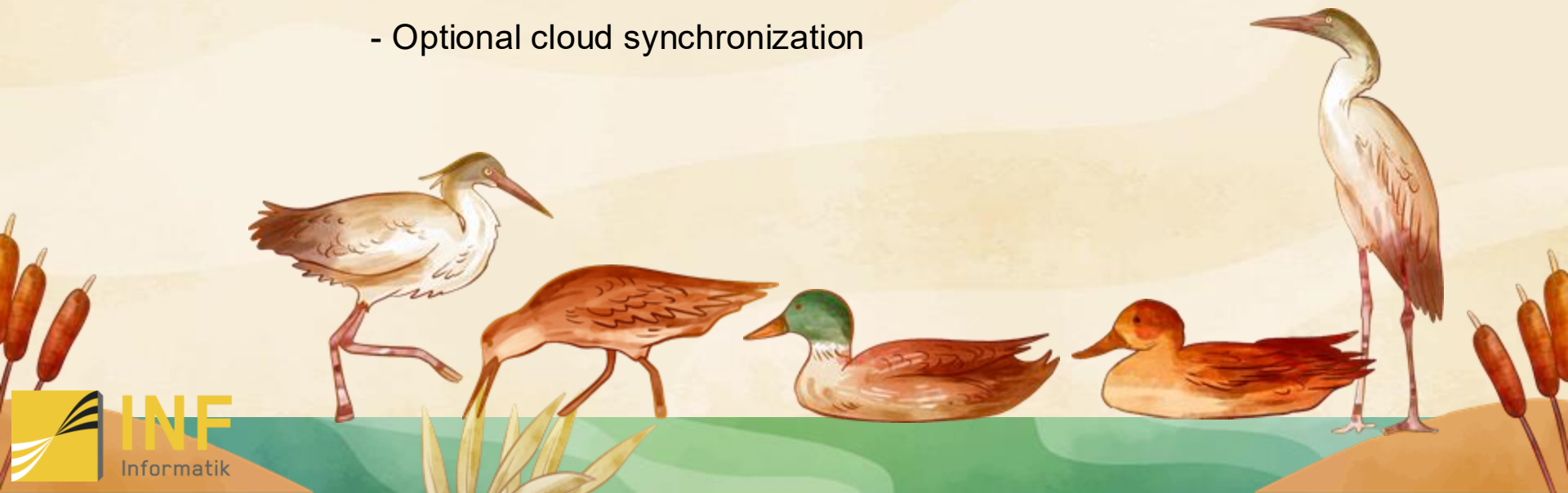


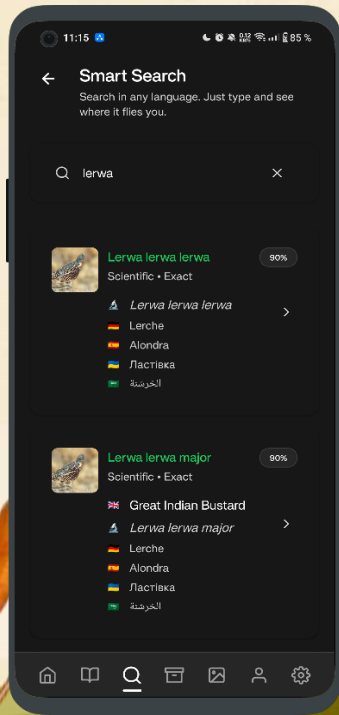


Fully local ! Optional bird sighting firebase storage



- Multi-modal capture (photo, audio, video, text)
- Draft management with auto-save
- Gallery organization and export
- Optional cloud synchronization





Smart Search with String Comparisson
for finding and comparing language
names

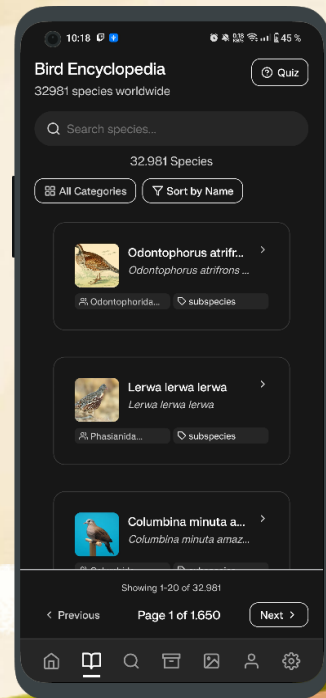
dynamic quiz for learning and
entertainment

All best-practice Cornell Lab Bird Facts in
local SQL Storage

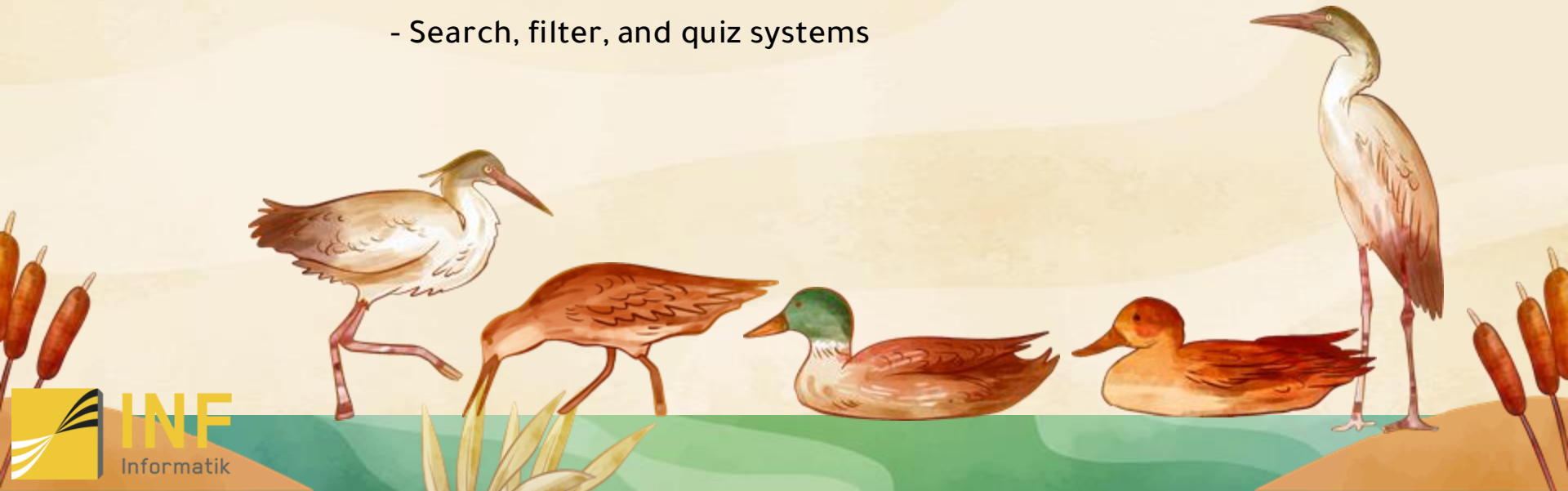
8k Bird Images from Wikimedia scraped,
stored locally, and mapped per bird
dynamically to its species

BirdyDex Encyclopedia

30k+ Birds
with names in
6 languages
and full
taxonomy
facts



- 33,000+ bird species database
- 6-language support (EN, DE, ES, FR, UK, AR)
- Wikimedia image integration (from local storage!)
- Search, filter, and quiz systems



App Language Support

01

English

Learn and Log Birds!

02

German

Lerne und dokumentiere
Vögel!

03

Spanish

¡Aprenda y documente las
aves!

04

Ukrainian

Вивчайте та
документуйте птахів!

05

French

Apprenez et documentez
les oiseaux !

08

Arabic

تعلم ووثق الطيور



Its deeemonstration timeee!



dive flight into the architecture



Framework

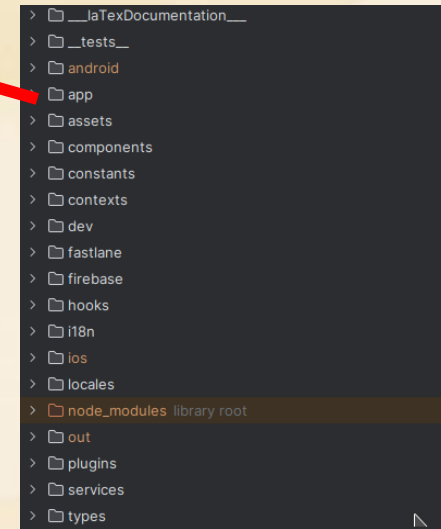
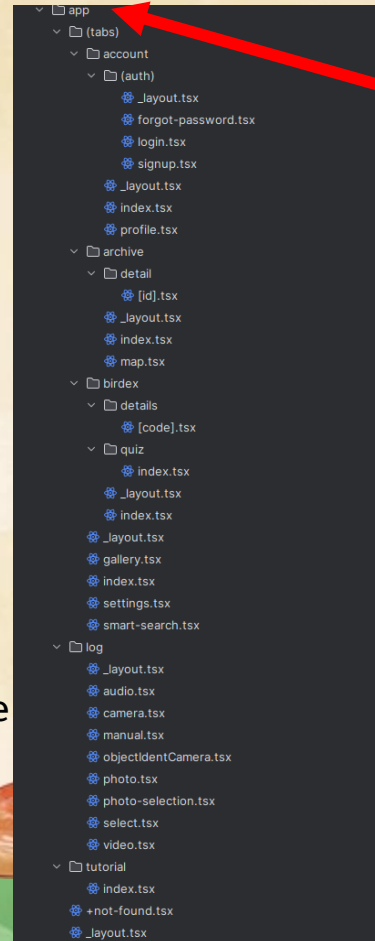
Technical Framework Stack

- React Native + Expo SDK - Cross-platform mobile development with TypeScript
- ML Integration - react-native-fast-tflite for BirdNET audio, MLKit for image detection
- Data Storage - SQLite (33,000+ species), Firebase Firestore (optional sync), AsyncStorage (drafts)
- UI Components - Custom themed system (ThemedView, ThemedText, ThemedPressable)
- Development - Metro bundler, EAS builds, custom dev builds for native ML modules



Layer & Navi Structure

- Primary Tabs (Expo Router file-based)
 - Home (feature dashboard)
 - BirDex (species encyclopedia)
 - Archive (observations & map)
 - Gallery (media management)
 - Account (authentication)
 - Settings (configuration)
- Logging Workflow (/app/log/)
 - Method selection → Manual/Photo/Audio/Video
 - AI Camera
 - Sequential capture flow with unified ML pipeline
 - Consistent back navigation pattern



Challenges

- Machine Learning needed Dev Client handling (Expo GO doesn't work)
- Correct and trying-to-be-universal handling of machine learning models in react native through none-deprecated packages (little documentation available)
- Permission handling, since rules have changed multiple times along the OS-Versions (less documentation, since older ones self-deprecated)
- File storage handling (and concurrent audio objects)
- 10K asset handling with non-dynamic loading
- Full local-first achitecture for the encyclopedia



Do you have any questions?

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Presentation by Marty Lauterbach

CREDITS: This presentation template was created by **Slidesgo**, and includes icons by **Flaticon**, and infographics & images by **Freepik**



LOGCHIPRY

fully local Ornithology

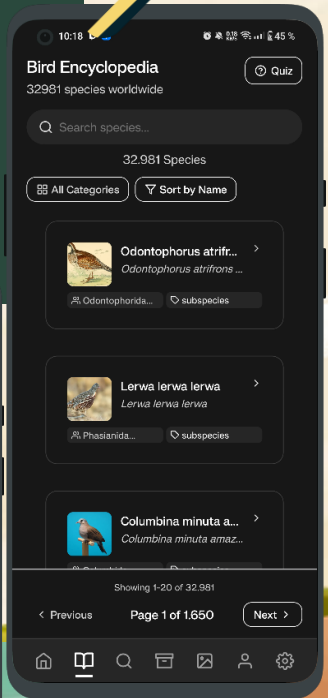
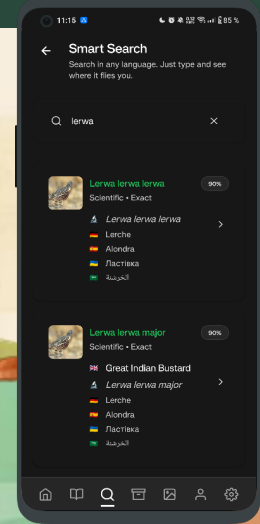
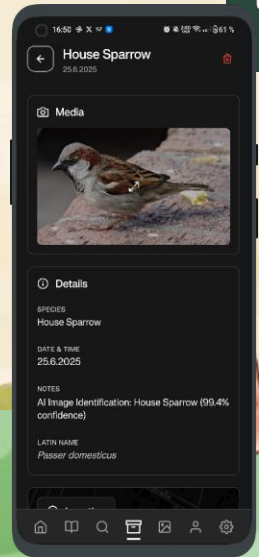
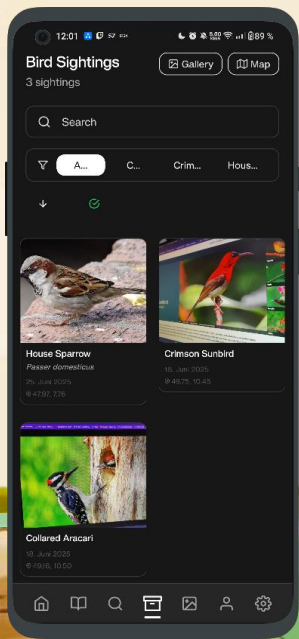
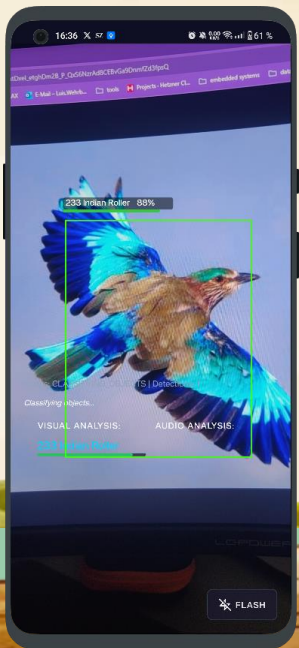
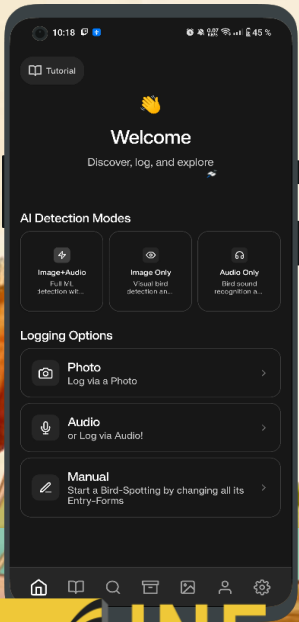


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